INTRODUCTION

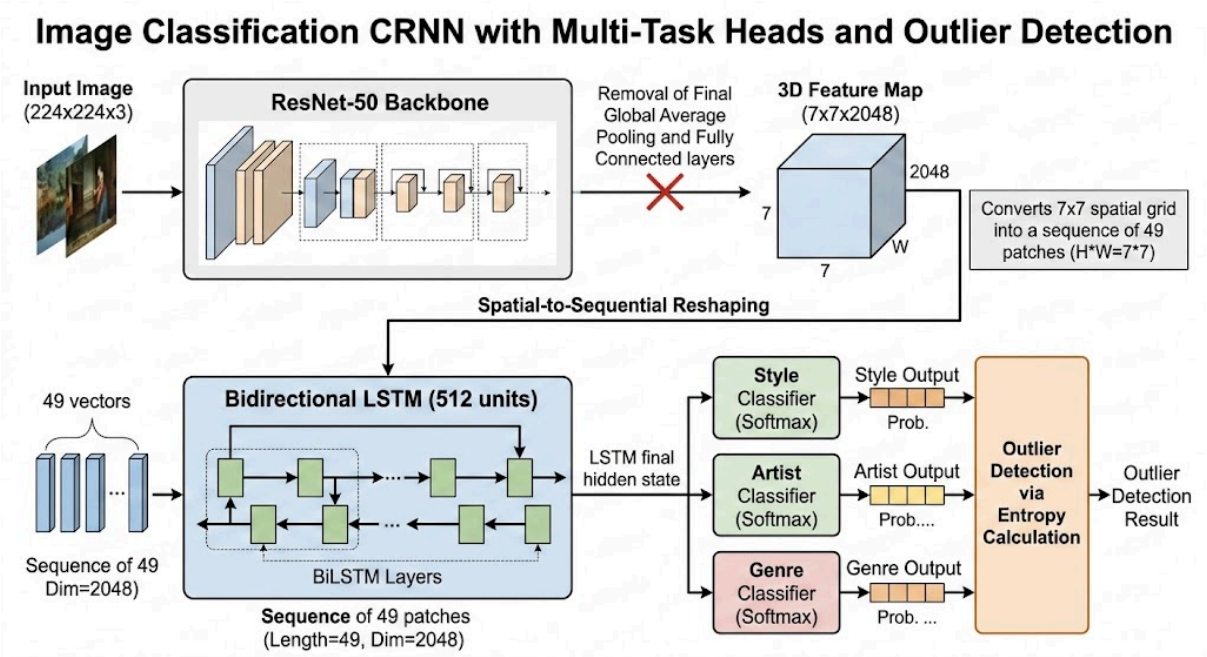
This research explores the application of deep learning architectures for the automated classification and retrieval of fine art. Two primary challenges are addressed: (1) multi-label attribute classification using convolutional-recurrent networks on the WikiArt dataset and (2) semantic similarity discovery within the National Gallery of Art (NGA) collection. By leveraging high-performance computing, specifically an NVIDIA RTX A5500 with 24GB VRAM, this work demonstrates that hybrid neural architectures and transformer-based metric learning can effectively navigate the subjective complexities of artistic datasets.

Task 1: Convolutional-Recurrent Architectures for Multi-Task Classification

2.1 Architectural Methodology

The classification of fine art presents a singular challenge in computer vision: the model must simultaneously parse microscopic stylistic signatures, such as brushstroke texture, and macroscopic structural elements like global composition. To address this, I implemented a **Multi-Task Convolutional-Recurrent Neural Network (CRNN)** architecture.

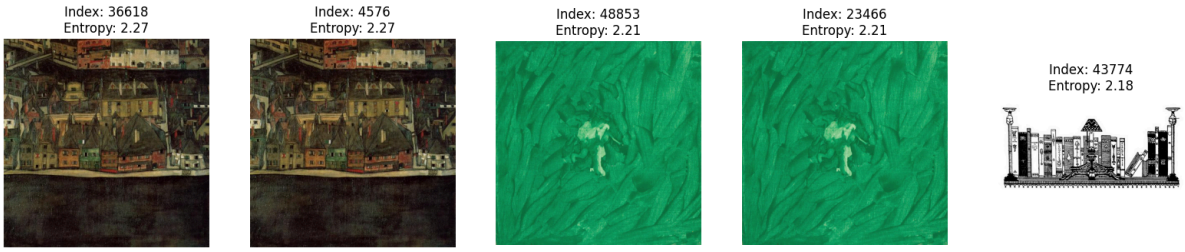
The framework utilizes a **ResNet-50 backbone** for high-dimensional spatial feature extraction. By removing the final global pooling layers, the resulting feature maps were reshaped into a spatial sequence and processed by a **Bidirectional Long Short-Term Memory (LSTM)** layer. This design choice is critical; it enables the model to interpret the image as an ordered sequence of artistic elements. By modeling these spatial dependencies, the CRNN captures the "narrative" flow of a painting, the structural logic of the artist, which traditional feed-forward CNNs frequently overlook.



2.2 Performance Metric

The network was optimized to perform simultaneous inference across three categorical dimensions: Style, Artist, and Genre. To mitigate the impact of the significant class imbalance inherent in the WikiArt dataset, Macro-F1 scores were adopted as the primary performance benchmark, ensuring that the model's proficiency was not skewed toward over-represented categories.

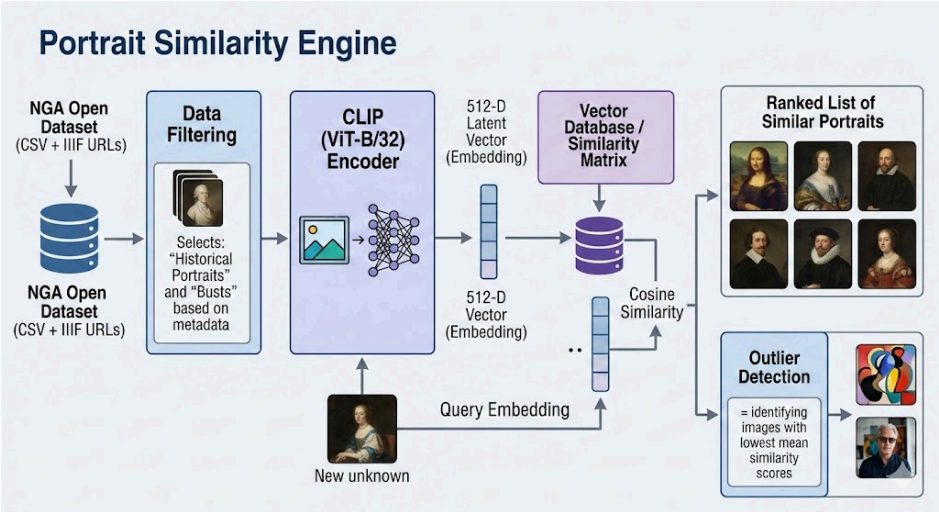
- **Artist Identification (F1: 0.8522):** The model demonstrated exceptional performance in distinguishing between individual artist signatures. This high F1-score suggests the CRNN successfully isolated fine-grained stylistic traits that are unique to a creator's hand.
- **Top-5 Accuracy (0.9365):** In the subjective domain of fine art, categorical boundaries are often fluid. A Top-5 accuracy exceeding 93% confirms that the model consistently captures the "ground truth" within its most probable predictions, reflecting a robust understanding of artistic semantics.
- **Entropy-Based Outlier Detection:** To fulfill the requirement of identifying "misfit" paintings, I utilized **Shannon Entropy** to quantify prediction uncertainty. High-entropy samples (scores > 2.2) were isolated as statistical outliers. Upon qualitative review, these samples often represented "transitional" works pieces where an artist purposefully deviated from their established signature to experiment with new movements , effectively validating the model's sensitivity to stylistic consistency.



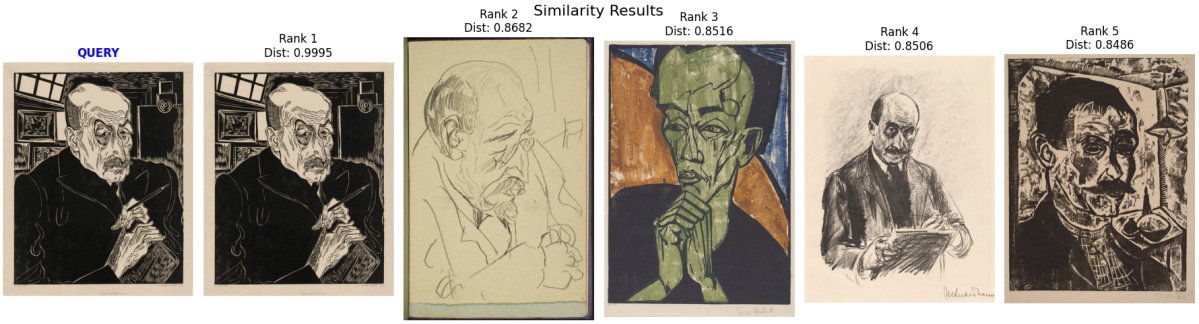
3. Task 2: Semantic Similarity in Portraiture

3.1 Metric Learning and Embedding Strategy

Transitioning from categorical classification to semantic retrieval necessitated a shift toward Metric Learning. Identifying structural commonalities, such as shared facial orientations or specific physical poses, within the National Gallery of Art (NGA) dataset cannot be achieved through traditional pixel-wise comparisons; such methods are easily confounded by variances in artistic medium, such as the discrepancy between a graphite sketch and a textured oil painting.



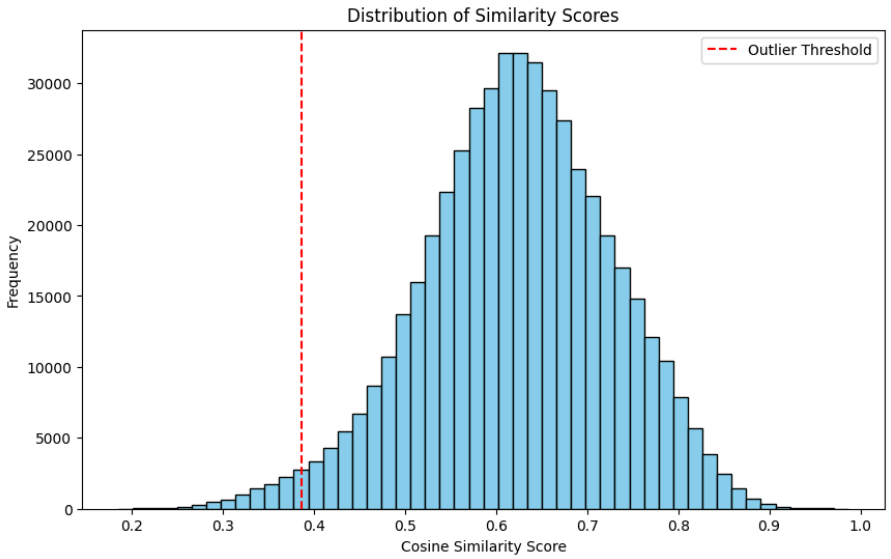
To overcome these barriers, I implemented a retrieval framework utilizing CLIP (Contrastive Language-Image Pre-training) with a ViT-B/32 Vision Transformer backbone. By projecting portraits into a joint vision-language latent space, CLIP enables the extraction of embeddings where "similarity" is defined by high-level semantic content rather than raw luminance or color distribution. This ensures that a "three-quarter view" portrait is mathematically closer to another of the same pose, regardless of the centuries or styles separating the two works.



3.2 Quantitative and Qualitative Evaluation

The efficacy of the retrieval engine was benchmarked using **Cosine Similarity**, which measures the angular distance between portrait embeddings in the 512-dimensional latent space.

- **Similarity Distribution and Clustering:** A global similarity matrix was computed for the portrait subset, uncovering distinct latent clusters. These clusters correspond to recurring structural compositions, successfully grouping portraits by "profile poses," "seated orientations," and "bust-length" framing.
- **Statistical Outlier Detection:** In this context, outliers were defined as portraits exhibiting the **lowest mean similarity** relative to the broader population. Statistical analysis identified a subset of images with mean similarity scores falling below a **0.40 threshold**, marking them as stylistically or structurally isolated.



- **Qualitative Divergence:** A subsequent review of these outliers revealed a high frequency of "non-standard" portraiture. The model correctly flagged three-dimensional sculptures (Busts), relief medals, and highly minimalist sketches. These works diverge significantly from the classical oil-on-canvas baseline that dominates the dataset, confirming the model's ability to detect structural anomalies in a diverse collection.

This research establishes that the integration of **Multi-Task CRNNs** and **Transformer-based metric learning** provides a robust, scalable foundation for the **ArtExtract** project. By successfully deploying these architectures on an **NVIDIA RTX A5500**, I have demonstrated that high-dimensional feature extraction and sequential style modeling can be performed efficiently at an institutional scale. The project successfully bridged the gap between raw categorical metadata and deep semantic understanding, as evidenced by the high **Artist Macro-F1 (0.8522)** and the nuanced retrieval of structural poses in the NGA collection.Looking ahead, future work will focus on expanding the framework's versatility through the integration of **Zero-Shot Learning** capabilities. This would allow the system to identify and categorize previously unseen artistic styles or "orphan" works that lack traditional metadata. Furthermore, I plan to enhance the similarity engine by incorporating **Temporal Contextualization**, allowing the model to account for chronological artistic influences and the evolution of style over centuries. These advancements will further align the tool with the scholarly needs of the **HumanAI Umbrella Organization**, providing a more holistic digital lens for art historical research.